

Week 2

Overview:

- Predicate Logic
- Kahoot
- Proof Patterns

DiskMath CHN E42

WhatsApp-Gruppe



Predicate Logic

Universe

Definition 2.10. A k -ary predicate²⁴ P on U is a function $U^k \rightarrow \{0, 1\}$.

Ex. " $x^2 \geq x$ "

$$U = \mathbb{R}, \quad P: \mathbb{R} \rightarrow \{0, 1\} \quad P(x) = 1 \Leftrightarrow x^2 \geq x$$

Universe is important!

If $U = \mathbb{Z}$ instead, then $P(x)$ is always 1.

- Universe $U \neq \emptyset$, e.g. $\mathbb{N}$, $\mathbb{Z}$, $\{1, 2, 3\}$, ...
 - Predicates
 - Functions $U^k \rightarrow U$ add $(x, y) := x + y$
 - Variables & Constants in U
e.g. $x = \underline{5} + \underline{2}$ $U = \mathbb{N} \checkmark$ $U = \{0, 1\} \times$
 constants $\in U$
 - logical operators $\wedge \vee \neg \rightarrow$ e.g. $(x > 3) \wedge (x < 5)$
 - Quantifiers $\forall, \exists$
e.g. $\exists x (x > 3)$ statement, not predicate
x is bounded, not free
(later in Chapter 6)
- Properties:

Properties:

The order of different quantifiers should not be changed:

Ex:

Universe : the set of all people

$M(x, y)$: x is the mother of y

- $\forall y \exists x : M(x, y)$: Everyone has a mother
 - $\exists x \forall y : M(x, y)$: There's someone who's the mother of everyone
- But $\forall x \forall y : P(x, y) = \forall y \forall x : P(x, y)$

But $\forall x \forall y P(x,y) \equiv \forall y \forall x P(x,y)$

$$\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$$

- Interpretations of formulas:

$$\underbrace{\forall x (P(x) \rightarrow Q(x))}_{\text{formula}}$$

Interpretation: $P(x) \stackrel{\text{def}}{=} x \text{ even } (2|x)$
 $Q(x) \stackrel{\text{def}}{=} x > 1$
 $U = \mathbb{N}$

Formula + Interpretation $\rightarrow$ Statement

$$\underbrace{\forall x (2|x \rightarrow x > 1)}_{\text{statement}}$$

- Equivalence and logical consequence in predicate logic

$F \equiv G$: For any interpretation of F and G,
 F and G are equivalent

$F \models G$: For any interpretation ...

Ex. $\forall x P(x) \models \exists x P(x)$

True, since $U \neq \emptyset$

- Useful Rules

- $\forall x P(x) \wedge \forall x Q(x) \equiv \forall x (P(x) \wedge Q(x))$

- $\exists x (P(x) \wedge Q(x)) \models \exists x P(x) \wedge \exists x Q(x)$

- $\exists x P(x) \wedge \exists x Q(x) \not\models \exists x (P(x) \wedge Q(x))$

$P(x)$: x hates DiskMath

$Q(x)$: x got a 6.0 in DiskMath

LHS: There is someone who hates DiskMath and someone who got a 6.0 in DiskMath

RHS: There is someone who hates DiskMath and got 6.0

- $\neg \forall x P(x) \equiv \exists x \neg P(x)$

- $\neg \exists x P(x) \equiv \forall x \neg P(x)$

- $\exists y \forall x P(x, y) \models \forall x \exists y P(x, y)$

- $\forall x \exists y P(x, y) \not\models \exists y \forall x P(x, y)$ - Example above

Proof Patterns

• Proof by Contradiction

Definition 2.17. A proof by contradiction of a statement S proceeds in three steps:

1. Find a suitable mathematical statement T .
2. Prove that T is false.
3. Assume that S is false and prove (from this assumption) that T is true (a contradiction).

Lemma 2.9. $(\neg A \rightarrow B) \wedge \neg B \models A$.

Exercise Sheet HS20

Let $x \in \mathbb{Q}$ and $y \in \mathbb{R} \setminus \mathbb{Q}$.

Prove by contradiction that $x+y$ is irrational.

Assume $x+y$ rational

$$\underline{x+y = \frac{m}{n}}$$

$$\underline{x = \frac{a}{b}}$$

$$\Rightarrow y = (x+y) - x = \frac{m}{n} - \frac{a}{b}$$

$\therefore y$ is rational

$$x = \frac{m}{n} \quad m, n \in \mathbb{Z}$$

... -2 -1 $\left[\begin{array}{c} 0 \ 1 \ 2 \ 3 \dots \\ \hline \mathbb{N} \end{array} \right]$

• Pigeonhole principle

Theorem 2.10. If a set of n objects is partitioned into $k < n$ sets, then at least one of these sets contains at least $\lceil \frac{n}{k} \rceil$ objects.⁴¹

n objects



k slots



$$\frac{n}{k} = 1.33\dots$$

$$\lceil \frac{n}{k} \rceil = 2$$

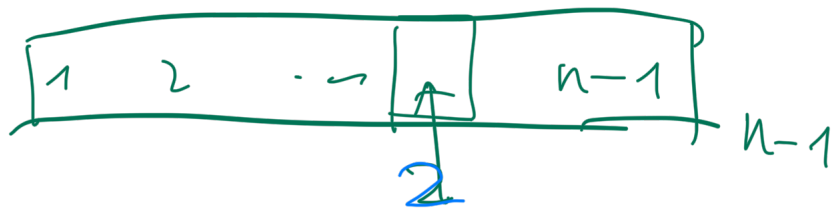
Exam HS 18

Task 5. Proof Patterns 4 Points

- a) $(\star \star)$ Let $a_1, \dots, a_n$ be any integers. Prove that there exists a sequence $a_i, a_{i+1}, \dots, a_j$ for some $1 \leq i \leq j \leq n$, such that the sum $a_i + a_{i+1} + \dots + a_j$ is divisible by n . (4 Points)

Hint: Use the pigeonhole principle.

Hint:
remainder 0



$$a_1 + \dots + a_5$$

$$a_1 + \dots + a_5 + \dots + a_{11}$$

$$\begin{matrix} n \\ - \end{matrix} \left\{ \begin{array}{l} a_1 / n \\ a_1 + a_2 / n \\ \boxed{a_1 + a_2 + a_3} \dots \\ a_1 + \dots + a_n \end{array} \right.$$

Kahoot

A: $F \text{ tautology} \Rightarrow G \text{ tautology}$

B: $F \models G$

$B \Rightarrow A$

$F \quad G \quad F \rightarrow G$

$F \text{ not tautology} \quad G \text{ not tautology}$

1	1
1	1
0	1
1	0

$F \rightarrow G \quad X$

1
1
1
0